

Pilot forecast

$P(\text{clear detection floor})$

low  high

+ uncertainty

GATE 1 · DETECTION

Measurement floor cleared
at the planned depth?

NO

YES

GATE 2 · IDENTIFICATION

Sibling alleles
separable?

YES

NO

ALLELE-LEVEL COMPARISON

Direction, PDS and DE fidelity;
model ranking with uncertainty

DETECTION-ONLY COMPARISON

Perturbation against reference only;
no allele-specific claim

MEASUREMENT-LIMITED

Report separately, not as model failure;
redesign or deepen, then re-pilot